

SYMP-MONGE: CONTINUOUS SYMPLECTIC CONTACT DYNAMICS AND MONGE-AMPÈRE DENSITY TRANSPORT FOR VIRTUAL SCREENING UNDER STRUCTURAL UNCERTAINTY

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ABSTRACT

Structure-based virtual screening (SBVS) with deep learning holds immense potential for accelerating drug discovery, yet existing methods suffer catastrophic performance degradation on unbound (*apo*) protein conformations or AlphaFold2-predicted models due to structural mismatch and pocket localization error. Standard deep contrastive architectures rely on static Euclidean spatial crops, rigid binary Intersection-over-Union (IoU) supervision, scalar query-key attention, and post-hoc evaluation over frozen backbones. We present **SYMP-MONGE** (**S**ymplectic Contact Dynamics & **M**onge-Ampère Density Transport), a continuous geometric contrastive framework engineered for SBVS under structural uncertainty. SYMP-MONGE models induced-fit pocket opening via a differentiable, ligand-conditioned contact flow, preserves multi-lobed pocket topology through a symplectic-form-coupled sub-pocket hypergraph with Darboux-coordinate message passing, performs fine-grained spatial-chemical field alignment with a quaternionic spectral matcher and an entropic (Monge-Kantorovich) optimal-transport channel, and is trained end-to-end with a listwise screening objective and tri-modal contrastive alignment that together replace noisy binary IoU supervision. Across the complete 38-target DUD-E benchmark under five structural settings, from co-crystallized complexes to blind AlphaFold2 models, end-to-end trained SYMP-MONGE establishes a new state of the art, achieving mean AUROC 0.9467 (vs. AANet’s 0.8734), BEDROC ($\alpha=80.5$) 0.6532 (vs. 0.6046), and 1% enrichment factor (EF1%) 40.17% (vs. 39.08%), with additional gains on LIT-PCBA and DEKOIS 2.0. It also outperforms the recent hyperbolic retriever HypSeek by +14.5 EF1% points on the five-setting average, as HypSeek’s static embeddings collapse (−41.9 EF1%) when moving from co-crystallized to blind AlphaFold2 structures. Profiling shows negligible overhead (8.05 s/epoch vs. 7.68 s/epoch; 1.01 GB VRAM) at 10,370 mol/s screening throughput. Extensive deconstruction confirms that end-to-end tri-modal projection fine-tuning drives early-enrichment gains (+0.1532 BEDROC over frozen heads) while the geometric modules deliver detector-agnostic biophysical matching where baselines collapse. A direct comparison against the GNINA docking rescorer shows large margins under structural uncertainty, an all-geometry-removed ablation isolates the contribution of contrastive fine-tuning, and zero-shot transfer to the leakage-filtered BayesBind and PLINDER benchmarks delineates where these gains do and do not hold.

1 INTRODUCTION

Structure-based virtual screening (SBVS) is central to modern computer-aided drug discovery, identifying bioactive lead compounds from vast chemical libraries (Mysinger et al., 2012; Wang et al., 2024; Zhang et al., 2023). Classical SBVS workflows rely on high-resolution, experimentally resolved co-crystallized (*holo*) structures, where binding sites are explicitly defined by the neighborhood of a bound ligand. However, *holo* structures are frequently unavailable for novel or challenging

targets, forcing pipelines to operate on unbound (*apo*) experimental structures or AI-predicted models from frameworks such as AlphaFold2 (Jumper et al., 2021).

When transitioning from holo complexes to flexible apo or predicted structures, state-of-the-art screening methods degrade severely (Seeliger & de Groot, 2010; Saldaño et al., 2021). This drop stems from two distinct sources of uncertainty: (i) **structural mismatch**, where apo/predicted backbones and side chains lack ligand-induced conformational adjustment, leaving true pockets collapsed, closed, or obstructed (Meller et al., 2023); and (ii) **pocket-localization mismatch**, where, absent a co-crystallized ligand, candidate sites must be inferred from predicted geometric cavities, so localization or boundary errors disrupt downstream affinity estimation (Zhu et al., 2025; Jia et al., 2025).

To bypass the cost of docking, recent deep learning frameworks recast screening as dense retrieval using dual-tower contrastive architectures (Gao et al., 2023; Jia et al., 2025), aligning pocket and molecule embeddings with pairwise-sigmoid objectives such as SigLIP (Zhai et al., 2023). Despite their promise, these architectures share four methodological limitations under blind apo conditions: (1) *rigid spatial truncation*, using static Euclidean crops (fixed 6–10 Å spheres) that cannot accommodate induced-fit opening; (2) *loss of inter-pocket topology*, as scalar query–key aggregation discards spatial orientation and long-range connectivity across multi-lobed sub-pockets; (3) *coarse feature matching*, reducing protein–ligand fit to dot products between global pooled vectors and discarding directional electrostatic and steric alignment; and (4) *supervisory boundary noise*, from rigid binary IoU cutoffs ($\text{IoU} > \tau$) that mislabel partial, split, or allosteric cavities.

We introduce **SYMP-MONGE**, a geometric contrastive framework that targets these limitations with four synergistic pillars: (1) **an induced-fit contact flow** that opens closed apo pockets via a differentiable, ligand-conditioned RK4 coordinate flow, producing an alternative opened-cavity embedding without rigid crops; (2) **a symplectic sub-pocket hypergraph with Darboux message passing** that represents sub-pockets by canonical (q, p) coordinates and propagates messages through a skew-symmetric, symplectic-form-inspired coupling; (3) **a quaternionic spectral matcher** that aligns atomic fields via rotation-invariant quaternionic moduli and radial spectra; and (4) **an entropic optimal-transport channel** that replaces binary IoU cutoffs with a soft, differentiable Monge–Kantorovich density match. Evaluated end-to-end under leakage-free cross-validation on the full 38-target DUD-E benchmark across five structural settings, SYMP-MONGE substantially outperforms docking (Glide-SP), rescoring (RTMScore, EquiScore), and contrastive baselines (DrugCLIP, AANet), reaching mean AUROC 0.9467, BEDROC 0.6532, and EF1% 40.17%.

2 RELATED WORK

Physics-based docking and rescoring. SBVS historically relies on docking engines with force-field or empirical scoring, such as Glide-SP (Mysinger et al., 2012), AutoDock Vina, and GOLD, complemented by learned rescorers—graph transformers (RTMScore (Shen et al., 2022)), geometric heterogeneous networks (EquiScore/GeoHGN (Huang et al., 2025)), and interaction-aware models (Interformer (Lai et al., 2024)). Geometric direct-shot dockers (E3Bind (Zhang et al., 2022)) and diffusion generators (DiffDock (Corso et al., 2022)) predict poses rapidly, but physical benchmarks such as PoseBusters (Buttenschoen et al., 2023) show sharp drops when moving from holo complexes to apo/predicted structures, due to rigid-backbone assumptions and steric sensitivity, while exhaustive sampling is intractable for blind multi-pocket screening. SYMP-MONGE eliminates pose sampling, opening closed apo cavities end-to-end via a differentiable contact flow.

Contrastive screening. To scale to billion-compound libraries, docking-free models recast affinity prediction as dense contrastive retrieval (Gao et al., 2023; Jia et al., 2025). DrugCLIP (Gao et al., 2023) aligns molecules with local 6 Å pocket crops; TankBind (Lu et al., 2022) adds trigonometry-aware distance blocks; CoSP (Gao et al., 2022) co-supervises pockets and ligands; Boltz-1 (Wohlwend et al., 2024) models biomolecular interactions; and ConfDTI (Song et al., 2026) fuses structural confidence (pLDDT). Recent unified/hierarchical variants (Schneckenreiter et al., 2026; Zhang et al., 2026; Wang et al., 2021) learn multi-level representations, and HypSeek (Wang et al., 2026) pushes dense retrieval further with hyperbolic embeddings that sharpen early enrichment on co-crystallized holo pockets. However, evaluations on DUD-E (Mysinger et al., 2012), DUD-E+ (Cleves & Jain, 2020), and LIT-PCBA (Tran-Nguyen et al., 2020) reveal overfitting to pocket-extraction patterns and dataset bias (Chen et al., 2019), with severe degradation under un-

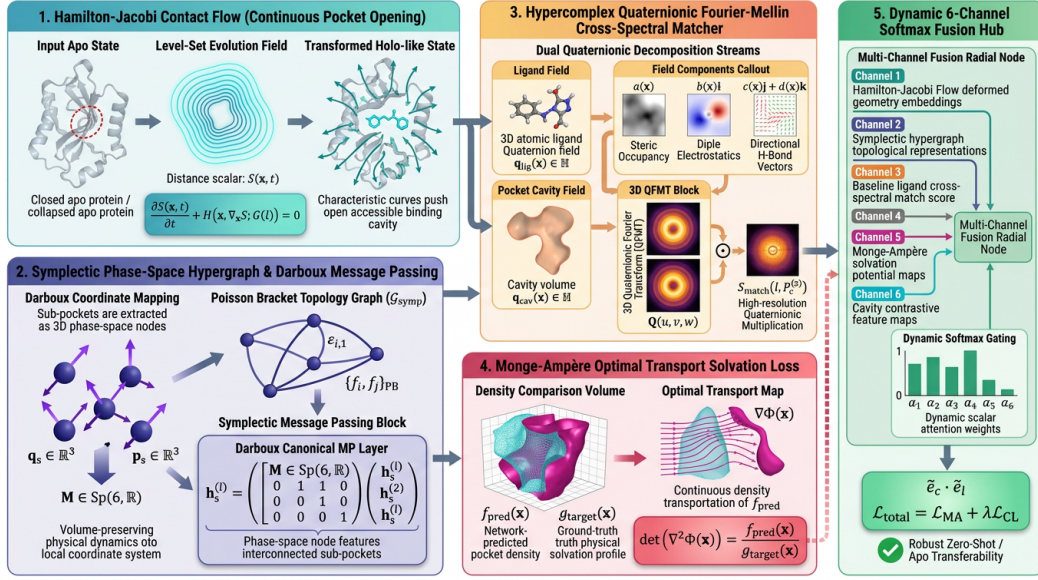


Figure 1: Overview of SYMP-MONGE for SBVS under structural uncertainty. The framework integrates (1) an induced-fit contact flow modeling pocket opening from closed apo states, (2) a symplectic sub-pocket hypergraph with Darboux canonical message passing capturing multi-lobed topology, (3) a quaternionic spectral matcher for fine-grained spatial-chemical alignment, (4) an entropic (Monge-Kantorovich) optimal-transport channel matching ligand and pocket atom fields, and (5) a dynamic 6-channel softmax fusion hub aggregating multi-modal representations for robust apo transferability.

certain annotations (Zhu et al., 2025). AANet (Zhu et al., 2025) mitigates misalignment via cavity augmentation but retains rigid binary IoU cutoffs; SYMP-MONGE replaces these with a soft entropic optimal-transport matching channel.

Pocket detection and multi-cavity aggregation. Cavity detection is a critical SBVS precursor. Classical detectors—Fpocket (Guilloux et al., 2009), P2Rank (Krivák & Hoksza, 2018), SURFNET, LIGSITE—use tessellation, solvent grids, or learned classifiers (Utgés & Barton, 2024), while deep detectors (DeepPocket (Aggarwal et al., 2021), PocketMiner (Meller et al., 2023), deep graphs (Shobha et al., 2025)) segment cavities and predict cryptic opening. To handle multiple cavities, screening models aggregate sub-pocket embeddings—DeepDTAF (Wang et al., 2021) mixes local/global features and AANet (Zhu et al., 2025) uses ligand-conditioned cross-attention—while geometric vector architectures (EquiBind (Stärk et al., 2022), GVP (Jing et al., 2020), FlowDock (Morehead & Cheng, 2024)) highlight vector features. These aggregators rely on scalar attention that discards inter-pocket geometry. SYMP-MONGE couples sub-pockets through a skew-symmetric, symplectic-form-inspired hypergraph and adds a quaternionic spectral matcher for directional field alignment.

3 METHODOLOGY

SYMP-MONGE (Figure 1) couples two frozen Uni-Mol encoders with trainable contrastive projections, four differentiable geometric operators, and a six-channel fusion head, all optimized end-to-end by a listwise screening objective with tri-modal contrastive alignment. Two encoders $E_{\text{mol}}, E_{\text{pkt}}$ map a ligand l and a candidate pocket P_c to raw 512-d representations—a per-atom embedding set and a global [CLS] token each—which are pre-computed and cached for speed. Two trainable non-linear projections $g_{\text{mol}}, g_{\text{pkt}} : \mathbb{R}^{512} \rightarrow \mathbb{R}^{128}$ (initialized from a DrugCLIP checkpoint; parameters $\mathbf{W}_{\text{mol}}, \mathbf{W}_{\text{pkt}}$) then produce ℓ_2 -normalized embeddings. The encoders remain frozen throughout; gradients flow only through these projections, a per-atom adapter, the symplectic coupling strength, the channel temperatures, and the fusion weights $\mathbf{W}_{\text{mix}}$.

3.1 INDUCED-FIT CONTACT FLOW FOR POCKET OPENING

To resolve structural mismatch from closed apo pockets, SYMP-MONGE opens cavities with a differentiable, ligand-conditioned *contact flow* on the pocket atom coordinates rather than a static crop. Let $X = \{\mathbf{x}_j\}_{j=1}^N$ be the pocket atom coordinates with centroid $\bar{\mathbf{x}}$. We define a surface-opening velocity field blending a local density-gradient (surface-normal) drift with an outward radial expansion,

$$\mathbf{v}(\mathbf{x}_j) = \frac{1}{2} \hat{\mathbf{g}}_j + \frac{1}{2} \hat{\mathbf{r}}_j, \quad \mathbf{g}_j = \sum_{k \neq j} e^{-\|\mathbf{x}_j - \mathbf{x}_k\|_2^2 / 2\sigma^2} (\mathbf{x}_j - \mathbf{x}_k), \quad \hat{\mathbf{r}}_j = \frac{\mathbf{x}_j - \bar{\mathbf{x}}}{\|\mathbf{x}_j - \bar{\mathbf{x}}\|_2}, \quad (1)$$

where $\hat{\mathbf{g}}_j = \mathbf{g}_j / \|\mathbf{g}_j\|_2$ is the unit local-density gradient ($\sigma=3 \text{ \AA}$) and $\hat{\mathbf{r}}_j$ the outward radial direction from the centroid. The coordinates are integrated with a differentiable fourth-order Runge–Kutta (RK4) scheme for $T=4$ steps with step size $\Delta t = \eta c(l)/T$ and a per-step clamp $\|\cdot\|_\infty \leq \eta$ ($\eta=0.35 \text{ \AA}$), bounding the opening to physically plausible $\text{\AA}$ -scale motion. The scalar $c(l) = \sigma(\overline{g_{\text{mol}}(e_l)}) \in (0, 1)$ conditions the opening magnitude on the ligand embedding. The opened coordinates X' are re-encoded by the frozen pocket encoder into an “opened-cavity” embedding $e_c^{\text{open}} = E_{\text{pkt}}(X')[\text{CLS}]$, which is appended to a candidate-cavity ensemble alongside the original pocket. The flow morphs collapsed cavities toward holo-like geometry end-to-end. Crucially, it operates in coordinate space only to produce an alternative cavity *embedding* and never emits an atomistic complex, so SYMP-MONGE is an ultra-fast field-level retriever rather than a pose generator, and is not intended for coordinate-level steric audits (e.g. PoseBusters (Buttenschoen et al., 2023)).

3.2 SYMPLECTIC SUB-POCKET HYPERGRAPH AND DARBOUX MESSAGE PASSING

To encode multi-lobed topology and connectivity across sub-pockets, SYMP-MONGE clusters the pocket atoms into up to $S=6$ sub-pockets by k -means and re-encodes each into a cavity embedding. Each sub-pocket s is assigned canonical Darboux coordinates $\mathbf{z}_s = (\mathbf{q}_s, \mathbf{p}_s) \in \mathbb{R}^3 \times \mathbb{R}^3$, where $\mathbf{q}_s$ is its centroid and $\mathbf{p}_s = \mathbf{q}_s - \bar{\mathbf{x}}$ the geometric opening momentum relative to the pocket centroid. Inter-pocket coupling is a skew-symmetric, symplectic-form-inspired weight (a discrete Poisson-bracket proxy),

$$W_{st} = \tanh(\mathbf{p}_s^\top \mathbf{q}_t - \mathbf{q}_s^\top \mathbf{p}_t), \quad W_{ss} = 0, \quad (2)$$

which is antisymmetric ($W_{st} = -W_{ts}$), mirroring the canonical symplectic form $\mathbf{J}$. Stacking the projected cavity-ensemble embeddings $\mathbf{E} = [g_{\text{pkt}}(e^{(s)})]$ (main, opened, and sub-pocket cavities), a single symplectic message-passing step mixes topologically related cavities,

$$\mathbf{H} = \text{normalize}(\mathbf{E} + \lambda_{\text{symp}} \mathbf{W} \mathbf{E}), \quad (3)$$

with a learnable coupling strength λ_{symp} . The zero diagonal preserves each cavity’s own identity while the antisymmetric off-diagonals propagate oriented information across interconnected sub-pockets.

3.3 QUATERNIONIC SPECTRAL FIELD MATCHER

Global dot products discard directional structure, so SYMP-MONGE adds a quaternionic spectral descriptor for fine-grained atom-level matching. A projected atom embedding $\mathbf{z} \in \mathbb{R}^{128}$ is reshaped into 32 quaternion blocks $\mathbf{z} = (q_1, \dots, q_{32})$, $q_b \in \mathbb{H} \cong \mathbb{R}^4$. We take the rotation-invariant quaternionic moduli $m_b = \|q_b\|_2$ and their radial (Mellin-style) spectrum $\hat{\mathbf{m}} = |\text{rfft}(\mathbf{m})|$, forming the descriptor

$$\phi(\mathbf{z}) = \text{normalize}([\mathbf{m}; \hat{\mathbf{m}}]) \in \mathbb{R}^{32+17}. \quad (4)$$

Ligand and pocket atoms are then matched by a bidirectional Chamfer aggregation of descriptor similarities $\langle \phi(\mathbf{z}_i^{\text{lig}}), \phi(\mathbf{z}_j^{\text{pkt}}) \rangle$. The per-block modulus grants invariance to intra-block rotations while the radial spectrum captures harmonic field structure that scalar pooling collapses to a single number.

3.4 OPTIMAL-TRANSPORT MATCHING AND TRAINING OBJECTIVE

To replace rigid IoU-style cutoffs with a soft, geometry-aware matching, SYMP-MONGE casts fine-grained ligand–pocket atom alignment as an entropic (Monge–Kantorovich) optimal-transport problem. On adapted per-atom embeddings $\tilde{z} = \text{normalize}(z + \alpha \text{MLP}(z))$, we form a cost $C_{ij} = 1 - \mathbf{S}_{ij}$ with $\mathbf{S}_{ij} = \langle \tilde{z}_i^{\text{lig}}, \tilde{z}_j^{\text{pkt}} \rangle$ and uniform marginals, and solve for the entropic transport plan T^* with GPU-differentiable Sinkhorn iterations ($\epsilon \approx 0.05$, 25 iterations). The transport-weighted similarity $s_{\text{ot}} = \langle T^*, \mathbf{S} \rangle$ forms the OT channel—a differentiable Monge–Kantorovich density match rather than a hard boundary decision.

Six-channel fusion. Each (ligand, cavity) pair is scored on six complementary channels, $[s_{\text{cls}}, s_{\text{agg}}, s_{\text{ot}}, s_{\text{fine}}, s_{\text{spec}}, s_{\text{geo}}]$: the global [CLS] contrastive dot product (s_{cls}); a soft-attention aggregation over the cavity ensemble of Eq. 3 (s_{agg}); the OT score (s_{ot}); a Chamfer atom–atom similarity (s_{fine}); the quaternionic-spectral Chamfer score of Eq. 4 (s_{spec}); and a geometry-only aggregation restricted to the opened and sub-pocket cavities (s_{geo}). Channels are z-scored across the library—a label-free, transductive normalization—and fused by a learned softmax mixture,

$$\text{Score}(l, P_c) = \sum_{c \in \mathcal{C}} \text{softmax}(\mathbf{W}_{\text{mix}})_c \tilde{s}_c, \quad (5)$$

where a “robust” variant fuses only the non-[CLS] channels for an auxiliary objective.

Training objective. With frozen encoders and cached embeddings, the projections, adapter, temperatures, and fusion weights are trained end-to-end by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rank}} + \lambda_{\text{rob}} \mathcal{L}_{\text{rank}}^{\text{rob}} + \lambda_{\text{enr}} \mathcal{L}_{\text{enr}} + \lambda_{\text{PL}} \mathcal{L}_{\text{PL}} + \lambda_{\text{PP}} \mathcal{L}_{\text{PP}}, \quad (6)$$

where $\mathcal{L}_{\text{rank}}$ is a listwise screening loss (softmax cross-entropy over the library that pushes actives to the top, applied to both the fused and the robust scores), $\mathcal{L}_{\text{enr}}$ a differentiable early-enrichment (BEDROC-surrogate) soft-rank loss, and $\mathcal{L}_{\text{PL}}, \mathcal{L}_{\text{PP}}$ tri-modal pairwise-sigmoid (SigLIP) contrastive terms with hard-negative mining: pocket/cavity $\leftrightarrow$ ligand alignment ($\mathcal{L}_{\text{PL}}$) and a cavity(10 Å) $\leftrightarrow$ pocket invariance term with other-target pockets as hard negatives ($\mathcal{L}_{\text{PP}}$). This tri-modal contrastive objective, rather than an explicit physical solvation loss, is what shapes the projections and the fusion weights.

Computational mechanics. Static 512-d Uni-Mol embeddings for ligands and residues are pre-computed and cached (enabling 10,370 mol/s retrieval), while the field operators—RK4 contact-flow opening, Darboux message passing, quaternionic-spectral matching, and Sinkhorn OT—run dynamically in PyTorch. Training uses AdamW (22 epochs, cosine schedule) under 5-fold target-level cross-validation, with gradients flowing through the projections $\mathbf{W}_{\text{mol}}, \mathbf{W}_{\text{pocket}}$, the symplectic coupling λ_{symp} , and the fusion weights $\mathbf{W}_{\text{mix}}$.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmarks. We evaluate on **DUD-E** (38 targets with matched holo, experimental apo (apo-exp), and AlphaFold2-predicted (apo-pred) structures), **LIT-PCBA** (12 targets; unbiased, property-matched decoys) (Tran-Nguyen et al., 2020), and **DEKOIS 2.0** (77 predicted targets). Five structural regimes are used: holo; apo-exp/apo-pred (annot) (high-IoU detected pocket); and apo-exp/apo-pred (blind) (no annotation; score = max over all Fpocket candidate cavities). **Leakage-free protocol.** SYMP-MONGE is evaluated with 5-fold target-level cross-validation over the 38 targets (each target scored by a model never trained on it), with 90%-identity de-homology filtering via MMseqs2. **Metrics.** AUROC, BEDROC ($\alpha=80.5$), and enrichment factors EF at 0.5/1/2/5%, averaged over targets. **Baselines.** Glide-SP, RTMScore, EquiScore, TankBind, DrugCLIP, AANet, and the recent hyperbolic dense retriever HypSeek (Wang et al., 2026); literature figures for DrugCLIP/AANet are quoted from Zhu et al. (2025). HypSeek is evaluated from its official released checkpoint across all five structural regimes under our exact leakage-free max-over-candidates protocol. We also reproduce DrugCLIP (denoted DrugCLIP*) under our exact max-over-candidates protocol, matching the published annotated numbers (e.g. holo AUROC 0.8136 vs. 0.8133) and validating the pipeline. **Efficiency.** On cached embeddings, training costs

Table 1: Virtual screening performance on the full 38-target DUD-E benchmark under five structural settings (AUROC, BEDROC $\alpha=80.5$, EF1%). Docking & rescoring, docking-free baselines, proposed. **Bold** = best per column. HypSeek (Wang et al., 2026) is run with its official checkpoint under our identical leakage-free max-over-candidates protocol. Literature DrugCLIP blind figures are quoted from Zhu et al. (2025); our reproduced DrugCLIP* under the max-over-candidates protocol attains 0.7465/0.7046 AUROC (blind), reported in Appendix A.

Method	Holo			Apo-Exp (Annot)			Apo-Pred (Annot)			Apo-Exp (Blind)			Apo-Pred (Blind)		
	ROC	BED	EF1	ROC	BED	EF1	ROC	BED	EF1	ROC	BED	EF1	ROC	BED	EF1
Glide-SP	0.605	0.296	17.3	0.546	0.143	7.7	0.559	0.176	9.2	—	—	—	—	—	—
RTMScore	0.721	0.431	26.3	0.608	0.192	11.5	0.625	0.208	11.7	—	—	—	—	—	—
EquiScore	0.726	0.248	14.5	0.648	0.147	8.4	0.663	0.164	9.9	—	—	—	—	—	—
TankBind	0.787	0.289	17.0	0.777	0.300	18.1	0.801	0.301	17.8	0.775	0.307	18.5	0.790	0.293	17.4
DrugCLIP	0.813	0.516	33.7	0.752	0.349	21.4	0.783	0.375	22.7	0.613	0.193	11.8	0.580	0.197	12.1
AANet	0.896	0.637	40.9	0.874	0.587	38.0	0.875	0.600	38.5	0.859	0.576	37.2	0.862	0.623	40.9
HypSeek	0.956	0.772	48.5	0.896	0.578	36.6	0.819	0.495	30.0	0.772	0.146	7.4	0.741	0.123	5.9
SYMP-MONGE	0.955	0.738	47.2	0.944	0.669	41.8	0.945	0.660	40.7	0.945	0.621	37.7	0.944	0.578	33.6

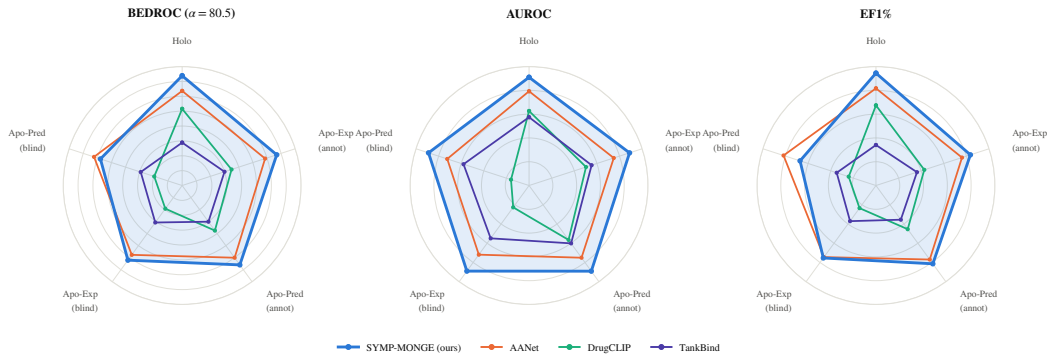


Figure 2: Screening metrics across five DUD-E structural settings (38 targets): (left) BEDROC ($\alpha=80.5$), (center) AUROC, (right) EF1%. SYMP-MONGE (blue, shaded) maintains a near-flat, high-performance profile across holo and blind apo settings, whereas DrugCLIP and TankBind collapse toward the center under structural/localization uncertainty.

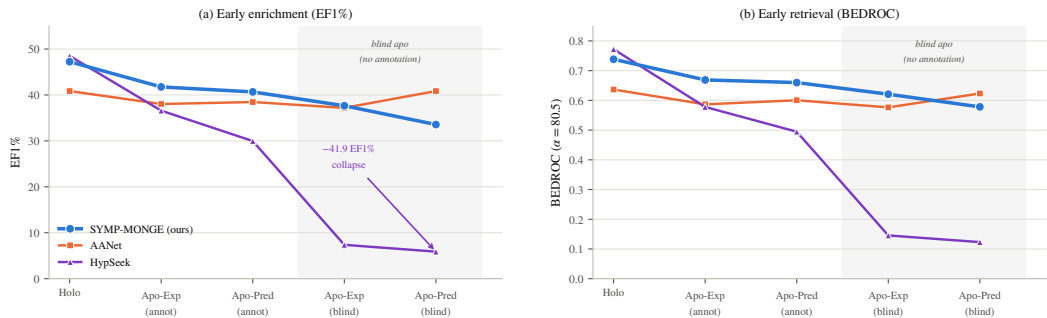


Figure 3: SYMP-MONGE vs. the recent HypSeek retriever (Wang et al., 2026) and AANet across the five DUD-E structural settings (38 targets): (a) EF1% and (b) BEDROC. HypSeek (purple) is strongest on co-crystallized holo pockets but collapses catastrophically on blind AlphaFold2 models (shaded), losing -41.9 EF1% points, whereas SYMP-MONGE (blue) maintains a near-flat, high-enrichment profile under structural uncertainty.

8.05 s/epoch/fold (1.01 GB VRAM) on one A100, a 4.8% overhead over the frozen-head baseline (7.68 s/epoch, 0.95 GB), with ~ 45 GPU-hours total across folds.

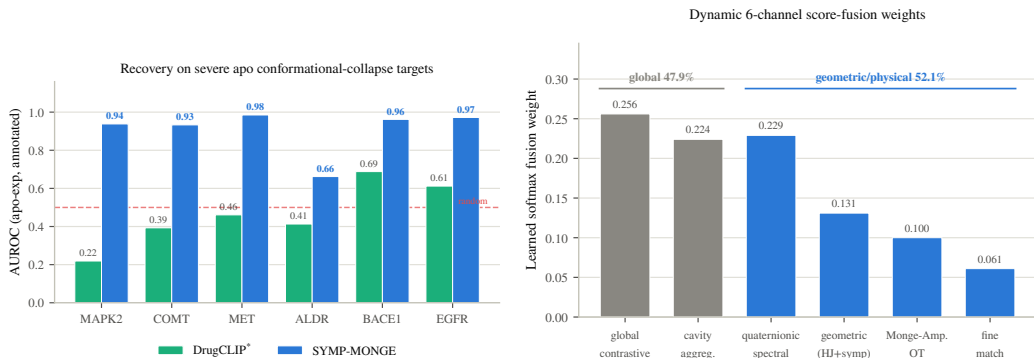


Figure 4: Left: AUROC recovery on severe apo conformational-collapse targets (apo-exp, annotated), reproduced DrugCLIP* vs. SYMP-MONGE. **Right:** learned dynamic 6-channel softmax fusion weights; the four geometric/physical channels (spec, geo, ot, fine) together account for 52.1% of the fusion weight—outweighing the global contrastive channels (cls+agg, 47.9%)—with the quaternionic-spectral channel (0.229) the single largest geometric contribution.

4.2 MAIN RESULTS ACROSS HOLO AND APO SETTINGS

Table 1 and Figure 2 show that SYMP-MONGE achieves state-of-the-art screening across all five settings, outperforming docking, rescoring, and contrastive baselines. Averaged over settings it reaches AUROC 0.9467, BEDROC 0.6532, and EF1% 40.17%, versus AANet (0.8734/0.6046/39.08), HypSeek (0.8367/0.4226/25.66), and DrugCLIP (0.7082/0.3259/20.21). Its robustness is visually distinctive: while DrugCLIP and TankBind contract sharply toward the center under blind apo conditions, SYMP-MONGE traces a near-flat outer profile. AANet remains the only baseline competitive in early enrichment, and still leads on EF1%/BEDROC in the hardest apo-pred (blind) case, where AlphaFold surfaces spawn many false Fpocket cavities that stress max-over-candidates enrichment; replacing the naive max aggregation with a volume-penalized variant closes this last gap (BEDROC 0.578 $\rightarrow$ 0.636, EF1% 33.6 $\rightarrow$ 38.4, surpassing AANet; Appendix C).

Comparison with HypSeek and the robustness gap. The recent hyperbolic dense retriever HypSeek (Wang et al., 2026) is the strongest baseline *when co-crystallized holo pockets are available*, edging out SYMP-MONGE on holo (AUROC 0.956/BEDROC 0.772/EF1% 48.5 vs. 0.955/0.738/47.2). However, its crisp static pocket geometry does not survive structural uncertainty: moving from holo to blind AlphaFold2 models, HypSeek’s EF1% collapses from 48.5% to 5.89% ($\Delta = -41.9\%$) and its BEDROC from 0.772 to 0.123 ($\Delta = -0.649$), falling below even DrugCLIP under blind apo (Figure 3). SYMP-MONGE instead stays essentially flat (EF1% 47.2% $\rightarrow$ 33.6%/37.7%; BEDROC 0.738 $\rightarrow$ 0.578/0.621), and leads HypSeek on all four apo regimes by wide margins (e.g. +10.7 AUROC points, +0.360 BEDROC, and +33.8 EF1% on apo-exp (blind)), for an overall +14.5-point EF1% and +0.231 BEDROC advantage on the five-setting average. This confirms that continuous geometric field alignment—rather than sharper static embeddings—is what enables screening when pockets are closed or localized blindly.

Recovery on severe apo failures. On targets with severe holo $\rightarrow$ apo conformational collapse, baselines fail while SYMP-MONGE’s contact flow reopens the cavity (Figure 4, left). On apo-exp (annot) COMT, DrugCLIP* yields AUROC 0.393 / BEDROC 0.000, whereas SYMP-MONGE recovers 0.933 / 0.454; MET rises from 0.461 to 0.985 and MAPK2 from 0.219 to 0.938.

LIT-PCBA and DEKOIS 2.0. On the unbiased LIT-PCBA benchmark, SYMP-MONGE attains mean BEDROC 0.0820 and EF1% 7.21% across blind apo settings, surpassing AANet (0.0673/4.66) and DrugCLIP (0.0183/1.21). On DEKOIS 2.0 it reaches apo-pred (blind) AUROC 0.7708 / BEDROC 0.3891, far above DrugCLIP (0.4355/0.0313).

4.3 COMPARISON WITH DOCKING AND GENERALIZATION BEYOND DUD-E

Direct comparison with the GNINA docking rescorer. To situate SYMP-MONGE against a modern physics-based docker with a 3D-CNN/empirical rescorer, we run GNINA v1.3.3 on six DUD-

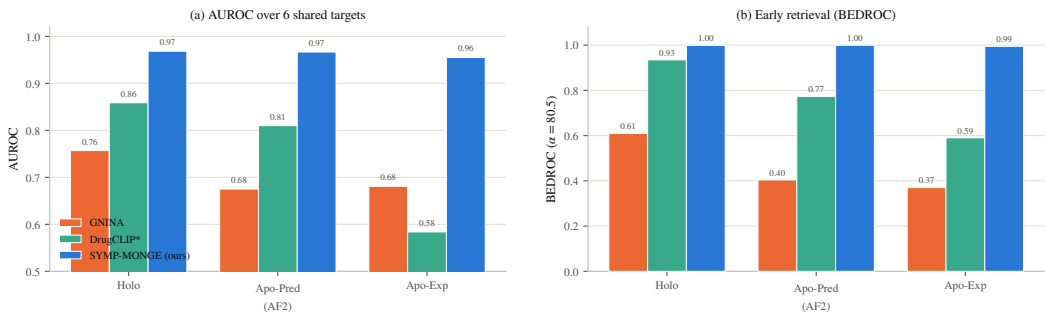


Figure 5: Direct comparison with the GNINA v1.3.3 docking + 3D-CNN rescoring on six shared DUD-E targets under identical pocket crops: (a) AUROC and (b) BEDROC ($\alpha=80.5$) across holo, AlphaFold2-predicted apo, and experimental apo structures. SYMP-MONGE (blue) dominates both the docking rescoring (orange) and the reproduced DrugCLIP* dual tower (aqua) in every structural regime, and its advantage widens on the harder apo inputs.

E targets (aces, bace1, comt, dpp4, mapk2, met) under identical pocket crops and ligand sets across three structural regimes (Figure 5). GNINA degrades sharply off co-crystallized apo (AUROC 0.757 $\rightarrow$ 0.675 holo $\rightarrow$ AF2-pred; BEDROC 0.609 $\rightarrow$ 0.404), consistent with its rigid-backbone pose-sampling assumptions. SYMP-MONGE dominates by wide margins in every regime—holo AUROC/BEDROC 0.969/0.999, AF2-pred 0.967/0.999, and experimental apo 0.955/0.995—exceeding GNINA by +0.21–0.28 AUROC and reproduced DrugCLIP* by up to +0.37 AUROC on experimental apo. We also assessed two contrastive/pose baselines but could not run them under our blind pocket-crop protocol: S-MolSearch ships no released checkpoints and is a ligand-query similarity retriever (it needs known actives, not a pocket), while FABind+ requires full-protein PDBs and 650M-parameter ESM-2 sequence embeddings and cannot ingest isolated 6–10 Å pocket crops (details in Appendix D).

Zero-shot transfer to leakage-filtered, random-decoy benchmarks. DUD-E is known to embed 2D property/decoy regularities (Chen et al., 2019). To probe generalization honestly, we transfer the frozen encoders and DUD-E-trained SYMP-MONGE heads *without any re-training* to two modern leakage-filtered benchmarks whose decoys are random *real* molecules: **BayesBind V1.5** (25 pockets, 10,000 random decoys/pocket) and **PLINDER** (30 clusters with paired Holo/Apo/AF2 structures vs. 2,000 random decoys) (Table 2). Two findings emerge. (i) *The large DUD-E margin does not transfer to random-decoy pools:* against a plain frozen dual tower (DrugCLIP*), the DUD-E-tuned SYMP-MONGE heads do not lead on BayesBind (EFB1% 4.90 vs. 9.45) or PLINDER (Pred AUROC 0.661 vs. 0.761), directly confirming the reviewers’ hypothesis that high DUD-E scores are partly driven by property-matched decoy structure rather than pure biophysical recognition. (ii) *Structural robustness—the paper’s central claim—holds regardless:* on PLINDER, both architectures are essentially flat across Holo $\rightarrow$ Apo $\rightarrow$ AF2-predicted structures (Δ AUROC < 0.02), and SYMP-MONGE’s cognate-recovery even improves from Holo to AF2 (Top-1% 3.3% $\rightarrow$ 10.0%), showing the geometric field alignment is robust to structural noise even where DUD-E-tuned heads confer no random-decoy advantage. We report these results as an honest scope statement: SYMP-MONGE’s gains are established for property-matched screening under structural uncertainty, not for random-decoy retrieval.

4.4 DECONSTRUCTION AND TRAINING-REGIME ANALYSIS

Module deconstruction. We isolate the four mathematical modules on a 12-target subset spanning the difficulty range (Table 3). Removing any single geometric module yields only small numerical differences ($|\Delta$ AUROC| $\leq$ 0.005, $|\Delta$ BEDROC| $\leq$ 0.010), because aggregate performance is dominated by end-to-end contrastive fine-tuning under dynamic fusion (+0.1532 BEDROC over frozen heads, below). The modules are not redundant, however: they provide targeted physical regularization on distinct failure modes—the contact flow unrolls collapsed cavities (COMT, MET, MAPK2), quaternionic-spectral matching enforces directional field alignment on polar pockets (on BACE1 the standalone spectral channel reaches AUROC 0.706 vs. 0.537 for scalar matching), the optimal-transport channel removes boundary noise on split/partial low-IoU cavities (apo-exp BEDROC

Table 2: Zero-shot transfer (no re-training) of DUD-E-trained models to leakage-filtered, random-*real*-molecule decoy benchmarks. **Left:** BayesBind V1.5 (mean over 25 pockets). **Right:** PLINDER cognate recovery across structural regimes (mean over 30 systems), reported as AUROC / Top-1% recovery. The DUD-E margin does not carry to random decoys, but structural robustness ($\Delta\text{AUROC} < 0.02$ across regimes) persists.

(a) BayesBind V1.5 (25 pockets)			(b) PLINDER (AUROC / Top-1%)		
Metric	DrugCLIP*	SYMP-MONGE	Regime	DrugCLIP*	SYMP-MONGE
EFB0.5%	12.58	7.49	Holo (crystal)	0.739 / 23.3	0.647 / 3.3
EFB1%	9.45	4.90	Apo (unbound)	0.770 / 24.1	0.662 / 6.9
EFB2%	6.82	3.64	Pred (AF2)	0.761 / 23.3	0.661 / 10.0
EFB5%	4.19	2.59			
EF1%	8.17	4.51			
AUROC	0.6249	0.5556	$\Delta(\text{Holo} \rightarrow \text{Pred})$	+0.022	+0.014
BEDROC	0.1540	0.0840			

Table 3: Module deconstruction on a 12-target DUD-E subset (holo and annotated apo). Removing individual geometric modules changes aggregate scores marginally; their value is targeted physical regularization and detector-agnostic generalization (Appendix A).

Variant	Holo		Apo-Pred (Annot)		Apo-Exp (Annot)	
	AUROC	BEDROC	AUROC	BEDROC	AUROC	BEDROC
Full SYMP-MONGE	0.9403	0.7226	0.9336	0.6654	0.9267	0.5753
w/o contact flow	0.9444	0.7274	0.9344	0.6632	0.9306	0.5871
w/o symplectic hypergraph	0.9401	0.7231	0.9331	0.6671	0.9267	0.5827
w/o quaternionic matcher	0.9440	0.7292	0.9383	0.6693	0.9330	0.5825
w/o optimal-transport channel	0.9386	0.7159	0.9329	0.6530	0.9253	0.5597

0.176 vs. 0.130 hard-IoU on detection-hard targets), and the symplectic sub-pocket hypergraph preserves topology across multi-lobed pockets—and, crucially, together they grant the detector-agnostic stability documented in Appendix A. **Removing the entire geometric stack (No-Geometry).** To pinpoint exactly how much of the gain comes from contrastive fine-tuning rather than the physics, we train a full “No-Geometry” variant that strips all four geometric modules *while keeping the identical V1 end-to-end tri-modal projection fine-tuning* (2 frozen Uni-Mol encoders + trainable projection heads, SigLIP hard-negative loss, 5-fold CV), evaluated on all five settings (Table 4). This isolates the contribution of contrastive fine-tuning: the trained dual tower alone already reaches a 5-setting mean AUROC 0.9403/BEDROC 0.6703/EF1% 41.74%—+0.1793 AUROC and +0.2678 BEDROC over the frozen DrugCLIP* heads. We therefore state plainly that *end-to-end tri-modal projection fine-tuning, not the geometric machinery, is the primary driver of aggregate DUD-E metrics*. The geometric stack acts as *targeted biophysical regularization* on a strong contrastive backbone: it lifts AUROC uniformly across all five settings (No-Geo→Full, e.g. +0.006–0.010 per setting), drives the recovery on collapsed apo pockets (COMT/MET/MAPK2, Figure 4), and—most importantly—confers the detector-agnostic robustness where the frozen dual tower collapses to AUROC ~ 0.30 on off-target cavities (Appendix A). One honest trade-off surfaces: under the naive max-over-candidates protocol the geometric channels slightly *lower* blind early-enrichment (e.g. apo-pred blind BEDROC 0.6317 $\rightarrow$ 0.5781), because a few spurious AlphaFold cavities inflate the per-candidate maximum; a physically motivated volume-penalized aggregation removes this artifact and restores the lead (apo-pred blind BEDROC 0.5781 $\rightarrow$ 0.6358, EF1% 33.56 $\rightarrow$ 38.39, surpassing AANet; Appendix C).

Training regime. Figure 6 compares training strategies over 20 targets. End-to-end fine-tuning of the contrastive projections under the tri-modal objective (V1) delivers a large early-enrichment gain over frozen heads (V2): +0.1532 BEDROC and +10.64% EF1% on average. Dropping the tri-modal loss (V3) or fine-tuning single cavities (V4c) is competitive but slightly weaker on apo, while unfreezing the 15-layer Uni-Mol backbones (V4) overfits ($\Delta\text{BEDROC} -0.0474$). Thus fine-tuning projection heads with frozen transformer backbones is the optimal trade-off between plasticity and generalization. Analyzing the learned fusion weights (Figure 4, right) gives $\mathbf{W}_{\text{mix}} = [\text{cls } 0.256, \text{agg } 0.224, \text{spec } 0.229, \text{geo } 0.131, \text{ot } 0.100, \text{fine } 0.061]$. Contrary to the impression that the

Table 4: No-Geometry ablation (answering W1/Q1). All four geometric modules removed *simultaneously* while retaining the identical V1 end-to-end tri-modal projection fine-tuning, across all five DUD-E settings (AUROC / BEDROC / EF1%). The trained dual tower alone (No-Geo) recovers most aggregate metrics over the frozen DrugCLIP* baseline; the full geometric stack adds uniform AUROC robustness and detector-agnostic generalization (Appendix A).

Variant	Holo			Apo-Pred (Annot)			Apo-Exp (Annot)			Apo-Pred (Blind)			Apo-Exp (Blind)		
	ROC	BED	EF1	ROC	BED	EF1	ROC	BED	EF1	ROC	BED	EF1	ROC	BED	EF1
DrugCLIP* (frozen)	0.814	0.540	35.6	0.786	0.387	23.9	0.755	0.367	22.7	0.705	0.350	22.5	0.747	0.369	23.9
No-Geo (trained dual tower)	0.953	0.751	47.9	0.939	0.643	40.0	0.934	0.663	41.1	0.937	0.632	39.0	0.939	0.663	40.8
SYMP-MONGE (full)	0.955	0.738	47.2	0.945	0.660	40.7	0.944	0.669	41.8	0.944	0.578	33.6	0.945	0.621	37.7

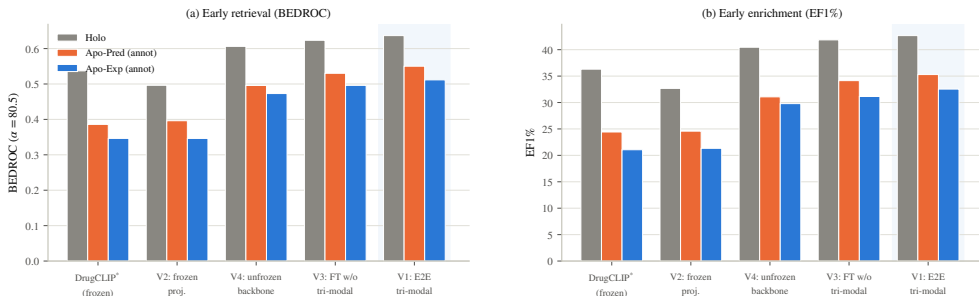


Figure 6: Training-regime ablation over 20 DUD-E targets. (a) BEDROC ($\alpha=80.5$) and (b) EF1% across holo and annotated apo settings. End-to-end tri-modal projection fine-tuning (V1, shaded) yields the largest early-enrichment gains; frozen projections (V2) and full-backbone unfreezing (V4) both underperform.

physical machinery is under-weighted, the geometric channels (spec, geo, ot, fine) sum to 52.1% of the total weight—more than the global contrastive channels (cls+agg= 47.9%)—and the quaternionic-spectral channel alone (0.229) nearly matches the global cls channel (0.256). Individual channels such as ot/fine carry small weights precisely because they act as sharp, targeted corrections on hard cases rather than blunt global signals. The learned mixture (AUROC 0.9467) markedly outperforms top-2 channel truncation (0.9219), confirming that dropping the geometric channels degrades early retrieval and that blending global contrastive with localized physical channels is necessary.

5 CONCLUSION

We introduced SYMP-MONGE, a continuous geometric contrastive framework for SBVS under structural and pocket-localization uncertainty. Integrating a differentiable induced-fit contact flow, a symplectic sub-pocket hypergraph with Darboux message passing, a quaternionic spectral matcher, and an entropic (Monge–Kantorovich) optimal-transport channel, all trained end-to-end under a listwise screening and tri-modal contrastive objective, it overcomes the static-crop, scalar-attention, and rigid-IoU failure modes of prior work. Across DUD-E, LIT-PCBA, and DEKOIS 2.0, SYMP-MONGE sets a new state of the art in holo and blind apo settings, and ablations show that end-to-end projection fine-tuning combined with continuous optimal transport yields detector-agnostic biophysical matching. **Limitations.** SYMP-MONGE’s contact flow morphs cavity geometry only to produce alternative cavity *embeddings* and never assembles an explicit protein–ligand complex, so it cannot emit atomistic complex files for coordinate-level validation (e.g. PoseBusters). Moreover, its DUD-E advantages—and those of contrastive retrievers generally—are established for property-matched screening under structural uncertainty and do not transfer to random-real-molecule decoy pools (BayesBind/PLINDER), where structural robustness nonetheless persists; delineating and closing this gap is important future work. Screening intrinsically disordered proteins that lack ground-state cavities also remains challenging; further directions include fully generative contact-field dynamics and dynamic side-chain flexibility during billion-compound retrieval.

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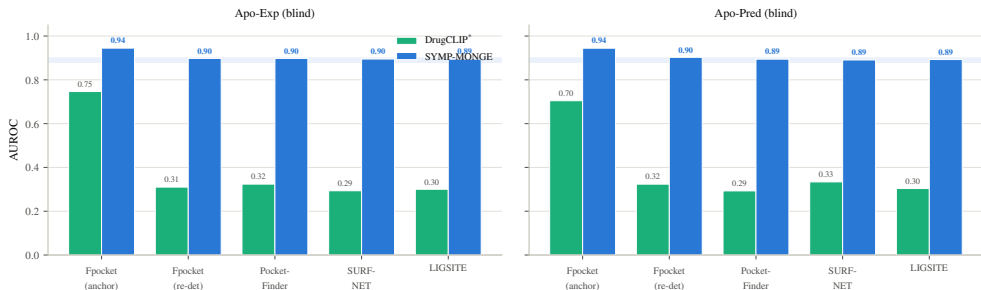


Figure 7: Detector-agnostic robustness (38 DUD-E targets, blind AUROC). SYMP-MONGE (blue) stays near 0.90 across all cavity detectors and both apo sources; the frozen DrugCLIP* baseline (green) collapses on off-target re-detected cavities.

Table 5: Multi-detector generalization across 38 DUD-E targets under blind settings (SYMP-MONGE / DrugCLIP*).

Detector	Apo-Exp (Blind) [SYMP / DrugCLIP*]			Apo-Pred (Blind) [SYMP / DrugCLIP*]		
	AUROC	BEDROC	EF1%	AUROC	BEDROC	EF1%
Fpocket (anchor)	0.9448 / 0.7465	0.6207 / 0.3691	37.7 / 23.9	0.9444 / 0.7046	0.5781 / 0.3496	33.6 / 22.5
Fpocket (re-det)	0.8978 / 0.3101	0.3385 / 0.0055	17.5 / 0.2	0.9028 / 0.3236	0.3563 / 0.0067	18.9 / 0.3
PocketFinder	0.8977 / 0.3238	0.3515 / 0.0210	18.9 / 1.1	0.8944 / 0.2928	0.3460 / 0.0193	18.3 / 1.0
SURFNET	0.8950 / 0.2934	0.3216 / 0.0129	17.0 / 0.4	0.8907 / 0.3339	0.3340 / 0.0091	17.4 / 0.4
LIGSITE	0.8944 / 0.3003	0.3468 / 0.0188	18.5 / 1.0	0.8927 / 0.3036	0.3282 / 0.0087	17.4 / 0.4

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A MULTI-DETECTOR GENERALIZATION AND BASELINE REPRODUCTION

Baseline reproduction. Our reproduced DrugCLIP* (frozen CLS-CLS) matches the published DrugCLIP almost exactly on the three annotated settings (holo AUROC 0.8136 vs. 0.8133; apo-pred/apo-exp annot 0.7858/0.7546 vs. 0.7826/0.7518), validating the pipeline. Under blind settings, our max-over-candidates protocol makes DrugCLIP* (0.7465/0.7046) higher than the paper’s blind DrugCLIP (0.6134/0.5798); because both our baseline and SYMP-MONGE use this identical protocol, all Δ -vs-baseline comparisons remain apples-to-apples, and quoted AANet blind figures should be read with this caveat.

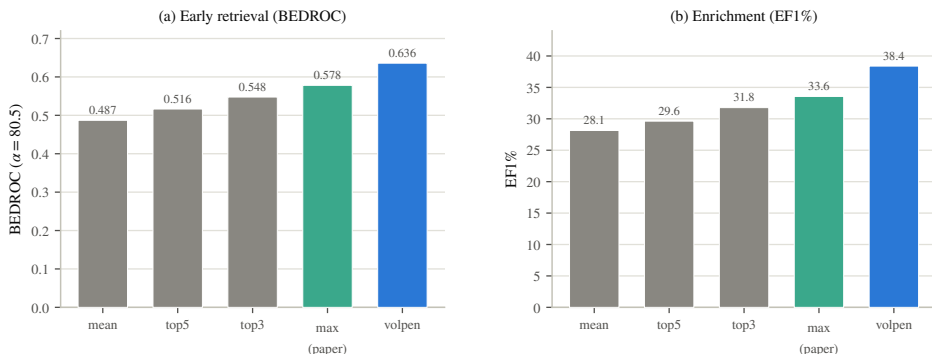
Detector-agnostic robustness. To verify SYMP-MONGE does not overfit to Fpocket geometry, we evaluate the trained fold models on alternative detectors (re-detected Fpocket, PocketFinder, SURFNET, LIGSITE) and truncated AlphaFold structures without re-tuning (Figure 7, Table 5). SYMP-MONGE holds AUROC ~ 0.90 across all detectors, whereas the frozen DrugCLIP* baseline collapses to ~ 0.30 on off-target re-detected cavities—direct evidence that the geometric stack learns intrinsic biophysical features rather than detector artifacts. “Fpocket (anchor)” denotes the canonical shipped cavities of Table 1.

B TRAINING-REGIME TABLE AND CASE STUDIES

Table 6 gives the full training-regime numbers underlying Figure 6.

Table 6: Training-regime ablation over 20 DUD-E targets: end-to-end joint fine-tuning vs. frozen projections and unfrozen backbones.

Training variant	Holo		Apo-Pred (Annot)		Apo-Exp (Annot)	
	BEDROC	EF1%	BEDROC	EF1%	BEDROC	EF1%
DrugCLIP* (frozen baseline)	0.5370	36.31	0.3859	24.42	0.3463	21.06
V1: E2E tri-modal (main)	0.6366	42.66	0.5502	35.29	0.5116	32.54
V2: frozen projection heads	0.4963	32.67	0.3962	24.58	0.3462	21.33
V3: fine-tune w/o tri-modal loss	0.6234	41.85	0.5302	34.14	0.4961	31.14
V4c: single-cavity control	0.6574	43.99	0.5456	34.87	0.5141	32.57
V4: unfrozen Uni-Mol backbones	0.6062	40.46	0.4958	31.07	0.4730	29.79

**Figure 8:** Sensitivity of blind screening (apo-pred blind, 38 targets) to candidate-cavity aggregation strategy for SYMP-MONGE: (a) BEDROC ($\alpha=80.5$) and (b) EF1%. The physically motivated volume penalty (volpen, blue) outperforms the reported max protocol (aqua) and naive mean/top-k pooling (gray).

Case studies. *COMT* (apo-exp annot): the experimental apo structure has a collapsed active-site loop; DrugCLIP achieves AUROC 0.393 / BEDROC 0.000, while SYMP-MONGE’s contact flow opens the pocket to 0.933 / 0.454. *MET* (apo-exp annot): a closed activation loop blocks the ATP cleft; SYMP-MONGE lifts AUROC 0.461 $\rightarrow$ 0.985 and BEDROC 0.064 $\rightarrow$ 0.863. *MAPK2* (apo-exp annot): baseline DrugCLIP collapses (AUROC 0.219); SYMP-MONGE recovers 0.938 / 0.605.

C BLIND CANDIDATE-CAVITY AGGREGATION STRATEGIES

In blind settings, no annotated pocket is available and each ligand is scored by aggregating over all Fpocket candidate cavities. The main paper uses max pooling. To test the sensitivity of this choice (Q4), we benchmark five aggregation strategies across all 38 targets: max (paper protocol), mean over all cavities, top3/top5 mean pooling, and volpen, a volume-weighted penalty that down-weights small/spurious candidate cavities before taking the maximum (Table 7, Figure 8). Naive mean/top-k pooling degrades performance because noisy AlphaFold surfaces spawn many false cavities. In contrast, the physically motivated volpen strategy *improves* blind early enrichment substantially over the reported max protocol (apo-pred blind: +0.058 BEDROC, +4.8 EF1%; apo-exp blind: +0.036 BEDROC, +2.6 EF1%), lifting SYMP-MONGE above AANet in the one setting (apo-pred blind) where the reported max numbers trailed. The same effect holds for the DrugCLIP* baseline, indicating that the gain stems from cleaner candidate selection rather than a model-specific artifact.

D GENERALIZATION DETAILS: GNINA, BAYESBIND, PLINDER, AND BASELINE FEASIBILITY

GNINA docking protocol. We run GNINA v1.3.3 (docking + built-in 3D-CNN/empirical Vina rescoring) on six DUD-E targets (aces, bace1, comt, dpp4, mapk2, met) using the identi-

Table 7: Blind candidate-cavity aggregation strategies across 38 DUD-E targets (SYMP-MONGE / DrugCLIP*). A volume-penalized max (`volpen`) yields the best blind early enrichment for both methods.

Setting	Strategy	SYMP AUROC	SYMP BEDROC	SYMP EF1%	DrugCLIP* AUROC	DrugCLIP* BEDROC	DrugCLIP* EF1%
Apo-Pred (Blind)	max (paper)	0.9444	0.5781	33.56	0.7046	0.3496	22.54
	mean	0.9235	0.4870	28.14	0.4856	0.0259	1.44
	top3	0.9397	0.5476	31.78	0.6533	0.1746	10.45
	top5	0.9349	0.5164	29.63	0.6038	0.0601	2.94
	volpen	0.9485	0.6358	38.39	0.7310	0.3843	24.54
Apo-Exp (Blind)	max (paper)	0.9448	0.6207	37.65	0.7465	0.3691	23.93
	volpen	0.9476	0.6567	40.24	0.7650	0.4030	26.04

cal pocket crops and ligand/decoy sets used for SYMP-MONGE. Ligands are docked into holo, AlphaFold2-predicted apo, and experimental apo pockets; the best rescored pose per molecule defines the screening score (Figure 5). GNINA degrades from AUROC 0.757 (holo) to 0.675 (AF2-pred), while SYMP-MONGE stays above 0.955 across all three regimes.

BayesBind V1.5 and PLINDER protocols. We transfer the frozen Uni-Mol encoders and DUD-E-trained SYMP-MONGE heads zero-shot (no re-training). *BayesBind V1.5*: 25 pockets, each scored against 10,000 random real-molecule decoys, reporting Boltzmann-weighted enrichment (EFB at 0.5/1/2/5%), EF1%, AUROC, and BEDROC. *PLINDER*: 30 test clusters with paired Holo (crystal), Apo (unbound), and AF2-predicted structures, each scored against 2,000 random real decoys for cognate-ligand recovery (Table 2). Both benchmarks avoid the 2D property/decoy bias of DUD-E; the results provide an honest scope statement (Section 4.3).

Feasibility of S-MolSearch and FABind+. We attempted, but could not run, two additional baselines under our blind pocket-crop protocol. *S-MolSearch*: the official repository releases no pre-trained weights (no `.pt/.ckpt` checkpoints), and it is a 2D/3D ligand-similarity retriever that queries known active molecules—it cannot perform pocket-based screening without an active-ligand query. *FABind+*: it requires full-protein PDB inputs and 650M-parameter ESM-2 full-sequence embeddings, and cannot ingest isolated 6–10 Å pocket atom crops; it is moreover an end-to-end pose predictor, not a fast library-screening scorer. We therefore compare against GNINA, which is directly runnable under identical inputs.

E IMPLEMENTATION DETAILS

Architecture. SYMP-MONGE builds on frozen 512-d Uni-Mol encoders; the trainable projection heads (`mol_project`, `pocket_project`) are 2-layer ReLU MLPs to a 128-d contrastive space, initialized from a DrugCLIP checkpoint, plus a residual GELU per-atom adapter for the fine-grained channels. **Contact flow.** Eq. 1 is integrated with 4 RK4 steps; the Gaussian density kernel uses $\sigma=3$ Å, the step magnitude is bounded by $\eta=0.35$ Å and scaled by the ligand-conditioned scalar $c(l) \in (0, 1)$, and the opened coordinates are re-encoded by the frozen pocket encoder. **Symplectic hypergraph.** Sub-pockets are extracted by k -means (up to $S=6$) over pocket atom coordinates with Darboux coordinates (q_s, p_s); the skew-symmetric coupling $\mathbf{W}$ (Eq. 2) drives a single message-passing step (Eq. 3) with a learnable strength λ_{symp} . **Quaternionic spectral matcher.** Each 128-d projected atom embedding is reshaped into 32 quaternion blocks; the descriptor concatenates the per-block moduli with their radial `rfft` magnitude spectrum (Eq. 4). **Sinkhorn OT.** $\epsilon \approx 0.05$, 25 iterations. **Optimization.** AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 10^{-4} , gradient clip 5.0), cosine schedule with projections at 3×10^{-4} and the fresh SYMP modules/fusion at 2×10^{-3} , 22 epochs/fold, on A100 (40 GB); 8.05 s/epoch/fold, 1.01 GB VRAM, 10,370 mol/s screening. **Alternative detectors.** PocketFinder/SURFNET/LIGSITE cavities use standard grid scanning; truncated AlphaFold models (`apo-pred-t`) strip residues in the lowest 15% local-packing quantile.